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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 20

Topics covered :

1. Vector Space of Functions and Orthogonality
2. Computation of Fourier Series Coefficients

- Let $\mathbb{V}$ of complex functions and Let $\phi(x)$ and $\psi(x)$ denote any two elements of the vector space. The inner product between the functions on the interval $[a, b]$ defined as

$$< \phi(x), \psi(x) >= \int_a^b \phi(x)\psi(x)^* dx \tag{1}$$

where $\psi(x)^*$ denotes the conjugate of the function $\psi(x)$. Note that $< \psi(x), \phi(x) >= (< \phi(x), \psi(x) >)^*$.

- The norm (L^2-norm) of the function on the interval $[a, b]$ is defined as

$$\|\phi(x)\| = \sqrt{< \phi(x), \phi(x) >} = \sqrt{\int_a^b |\phi(x)|^2 dx} \tag{2}$$

For a real-value signal $\phi(x) : \mathcal{R} \to \mathcal{R}$, the L^2-norm of the function on the interval $[-\infty, \infty]$ is related to the energy (E) of the signal [1].

$$E = \int_{-\infty}^{\infty} |\phi(x)|^2 dx = \|\phi(x)\|^2 \tag{3}$$

- Two functions $\phi(x)$ and $\psi(x)$ are orthogonal on $[a, b]$ if their inner product is zero i.e., $< \phi(x), \psi(x) >= 0$.

- Let $\{\phi_1(x), \cdots \phi_n(x)\}$ denote a set of orthogonal functions on the interval I. The linear combination of the orthogonal functions is given by

$$\sum_{i=1}^{n} C_i \phi_i(x) \tag{4}$$

- In Fourier Analysis, we typically work with the set of functions $\{\phi_n(x)\}_{n=-\infty}^{\infty}$, where $\phi_n(x) = e^{jn\omega_0 x}$, with n being an integer and $\omega_0 = \dfrac{2\pi}{L}$ representing the fundamental frequency. It can easily be shown that the the functions in the set are orthogonal i.e., $\phi_m(x)$ and $\phi_k(x)$ are orthogonal $\forall m \neq k$ over the interval $[0, L]$.

$$\begin{aligned} < \phi_m(x), \phi_k(x) > &= \int_0^L e^{jm\omega_0 x} \left(e^{jk\omega_0 x}\right)^* dx = \int_0^L e^{j(m-k)\omega_0 x} dx \\ &= \frac{1}{j(m-k)\omega_0} \left. e^{j(m-k)\omega_0 x} \right|_0^L = 0 \end{aligned} \tag{5}$$

On the other hand, the norm of every function $\phi_n(x)$ on $[0, L]$ in the set is given by

$$|\phi_n(x)| = \sqrt{< \phi_n(x), \phi_n(x) >} = \sqrt{\int_0^L e^{jn\omega_0 x} \left(e^{jn\omega_0 x}\right)^* dx} = \sqrt{\int_0^L dx} = \sqrt{L} \tag{6}$$

[1] This is true even for complex-valued signal but with minor difference in the definition of energy

One approach to understanding the Fourier series is as follows: Given a periodic function $g(x)$ with period L, the goal is to determine the coefficients C_n, referred to as **Fourier Coefficients**, such that the function $g(x)$ can be expressed as a linear combination of the set of functions $\{\phi_n(x)\}_{n=-\infty}^{\infty}$ i.e.,

$$g(x) = \sum_{n=-\infty}^{\infty} C_n \underbrace{e^{jn\omega_0 x}}_{\phi_n(x)} \tag{7}$$

Equation (7) is referred to as the **Fourier Synthesis** equation.

- **Determination of Fourier Coefficients:** One way to estimate the Fourier Coefficients is by using the orthogonality property set of functions $\{\phi_n(x)\}_{n=-\infty}^{\infty}$. Taking the inner product with $\phi_k(x)$ on both sides of (7) results in

$$< g(x), \phi_k(x) >= \sum_{n=-\infty}^{\infty} < C_n\phi_n(x), \phi_k(x) >= \left(\sum_{n=-\infty, n\neq k}^{\infty} < C_n\phi_n(x), \phi_k(x) > \right) + < C_k\phi_k(x), \phi_k(x) >= C_k L \tag{8}$$

Thus, the Fourier coefficient C_k can be computed as

$$C_k = \frac{1}{L} < g(x), \phi_k(x) >= \frac{1}{L}\int_0^L g(x)e^{-jk\omega_0 x}dx = \frac{< g(x), \phi_k(x) >}{< \phi_k(x), \phi_k(x) >} \tag{9}$$

Equation (9) is known as the **Fourier Analysis** equation. It is essential to recognize that the derivation of (9) assumes the validity of (7), an assumption that will be addressed in during the discussion on the convergence of Fourier series.

- For **real-valued functions**, the Fourier coefficients C_k and C_{-k} are related as

$$C_{-k} = \frac{1}{L}\int_0^L g(x)e^{-j(-k)\omega_0 x}dx = \left(\frac{1}{L}\int_0^L g(x)e^{-j(k)\omega_0 x}dx\right)^* = C_k^* \tag{10}$$

- **Projection of a function onto an orthogonal set**[2]: Let $\{\phi_1(x), \cdots \phi_n(x)\}$ represent a set of orthogonal functions defined on the interval I. Let $g(x)$ be a function defined on I. The objective is to determine the coefficients $c_1, \cdots, c_n$ of the linear combination such that

$$D = \left\| g(x) - \sum_{i=1}^{n} c_i\phi_i(x) \right\|^2 \tag{11}$$

is minimized. Note that D represents the least square error between the function and its approximation by the linear combination of the orthogonal functions.

(Stated without proof): The minimum value of D is attained uniquely when

$$c_i = \frac{< g(x), \phi_i(x) >}{\|\phi_i(x)\|^2} \tag{12}$$

[2]This concept is more intuitive if the reader is familiar with the projection of a vector onto a subspace

(Refer to [1] for proof). Further, the linear combination

$$p(x) = \sum_{i=1}^{n} \underbrace{\frac{< g(x), \phi_i(x) >}{\|\phi_i(x)\|^2}}_{c_i} \phi_i(x) \tag{13}$$

is called the projection of $g(x)$ onto the orthogonal set $\{\phi_1(x), \cdots \phi_n(x)\}$.

- The Fourier synthesis equation represents the projection of a periodic function $g(x)$ onto the orthogonal set of complex exponentials.

- **Relation between Complex Fourier Coefficients and Trigonometric Fourier Coefficients**: The Fourier coefficients a_n and b_n of the Trigonometric Fourier series

$$a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(n\omega_0 x\right) + b_n \sin\left(n\omega_0 x\right)\right] \tag{14}$$

can be computed from the Fourier series in complex exponential form. Expanding and rearranging the complex Fourier synthesis equation results in

$$g(x) = C_0 + (C_1 + C_{-1})\cos(\omega_0 x) + j\,(C_1 - C_{-1})\sin(\omega_0 x) + \cdots + (C_k + C_{-k})\cos(k\omega_0 x) + j\,(C_k - C_{-k})\sin(k\omega_0 x) + \cdots \tag{15}$$

Comparing (15) and (14) gives

$$a_0 = C_0,\ a_k = C_k + C_{-k}\ \forall k \neq 0, \text{ and } b_k = j\,(C_k - C_{-k})\ \ \forall k \neq 0 \tag{16}$$

For real values functions $C_{-k} = C_k^*$ and as a result, the relation between the trigonometric and complex exponential Fourier coefficients is given by

$$a_0 = C_0,\ a_k = 2\text{Real}\{C_k\}\ \forall k \neq 0, \text{ and } b_k = 2\text{Imag}\{C_k\}\ \forall k \neq 0 \tag{17}$$

References

[1] M. A. Pinsky, *Partial Differential Equations and Boundary-Value Problems with Applications*, 3rd ed. Providence, RI: American Mathematical Society, 2011.